

Power Functions

SUPPORTING SKILLS LEGEND

Student learning objectives for this unit

I — Introduce

D — Deepen

A — Apply

R — Reinforce

- ☐ 1. I can **identify** a power function and interpret the parameters in $y = kx^p$.

RELATED SKILLS

- ☐ D Interpret the coefficient and exponent in a power function.
- ☐ D Represent and interpret a power-variation relationship.
- ☐ R Recall and correctly use power-function terms including coefficient, exponent, direct variation, and inverse variation.

- ☐ 2. I can **construct** a power function from two data points or a proportional relationship.

RELATED SKILLS

- ☐ D Determine a power function from two points or a proportional relationship.
- ☐ A Isolate a specified variable in an equation or formula.

- ☐ 3. I can **relate** the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.

RELATED SKILLS

- ☐ D Determine power-function behavior as the input approaches zero or positive or negative infinity.
- ☐ D Relate the coefficient and exponent of a power function to its graph shape.
- ☐ A Connect parameters in a formula to features of its graph.

- ☐ 4. I can **recognize and model** direct, inverse, and other power-variation relationships.

RELATED SKILLS

- ☐ D Represent and interpret a direct-variation relationship.
- ☐ D Represent and interpret a power-variation relationship.
- ☐ D Represent and interpret an inverse-variation relationship.

- ☐ 5. I can **solve** equations involving integer or rational powers.

RELATED SKILLS

- ☐ D Interpret numerator and denominator roles in a rational exponent.
- ☐ A Use inverse operations to maintain an equivalent equation.

☐ **6. I can **build or select** a power-function model for a context and interpret its parameters.**

RELATED SKILLS

- I** Select a function family whose behavior fits a table, graph, or context.
 - D** Explain a model parameter in the language of its context.
 - A** Determine a power function from two points or a proportional relationship.
 - A** Restrict a model to meaningful contextual inputs.
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☐ **7. I can **analyze** power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.**

RELATED SKILLS

- I** Analyze the domain, graph shape, and behavior near zero of a power function with a fractional exponent.
 - A** Interpret numerator and denominator roles in a rational exponent.
 - A** Require an even-root radicand to be nonnegative when determining domain.
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